



2026 Chantilly Math Competition

Elementary School Division - Individual Round

6TH, 7TH, AND 8TH GRADERS **MUST** COMPETE IN THE MIDDLE SCHOOL DIVISION!

Competition Instructions

1. Do not open the test until instructed.
2. No computational aids other than pencil/pen are permitted.
3. Write answers in designated boxes on answer sheet.
4. All answers must be nonnegative integers: $\{0, 1, 2, \dots\}$.
5. Points per question range from 9 to 16.
6. The test consists of 16 short-answer questions to be solved in 60 minutes.
7. The final question will be used for tie-breaking.
8. There is no penalty for incorrect answers.
9. You will be graded on your answers—not your work.
10. If you have any questions, ask your proctor.

11. **Notation/Definitions:**

- The dot $\cdot$ denotes multiplication: $a \cdot b = a \times b$.
 - $n!$ denotes the factorial of n : $n! = n \times (n - 1) \times \dots \times 1$.
 - Two integers a and b are *relatively prime* if and only if $\gcd(a, b) = 1$.
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1. (9 pts) What is the mean of the numbers 15, 30, 45, 60, and 75?
2. (9 pts) Nischal takes a class with 7 tests. Nischal's average score on the first 6 tests is 76. If Nischal needs his average score on all 7 tests to be at least 67 to pass, what is the minimum possible score Nischal could get on his 7th test and still pass?
3. (10 pts) Which integer is the closest to the circumference of a circle with radius 9?
4. (10 pts) Farmer John only has chickens and cows on his farm. There are 67 animals on the farm. These animals have 200 legs altogether. How many chickens are on the farm?
5. (11 pts) A bag has 6 red balls, 7 green balls, and 8 blue balls. What is the fewest number of balls that need to be removed from the bag so that there are more red balls than balls that aren't red?
6. (11 pts) Boris is buying a sweater. The price tag says \$100 on it. Boris has a coupon that gives him a 6% discount, but there is also a 7% sales tax. After the coupon and sales tax are both applied, by how many cents does the price change?
7. (12 pts) A plane is scheduled to depart from City A at 5 PM and arrive at City B at 8 PM, traveling at an average speed of 550 mph. If the plane is instead delayed by 30 minutes, what is the new average speed required to arrive at City B on time, in mph?
8. (12 pts) Darren has a spherical soap. After 234 showers, the soap is a sphere with 40% of its original radius. Assuming every shower uses the same amount of soap, how many more showers will the soap last?
9. (13 pts) Find the smallest odd prime that is 2 more than a multiple of the sum of its digits.
10. (13 pts) Alice writes the positive integers from 1 to n inclusive on a whiteboard. Bob erases some (but not all) of the positive integers, so the average of the remaining integers is exactly 6.7. What is the minimum possible value of n ?
11. (14 pts) Compute the smallest positive integer divisible by 999 with all digits differing.
12. (14 pts) If

$$\frac{20}{23} = \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}}$$

for positive integers a , b , c , and d , compute $1000a + 100b + 10c + d$.

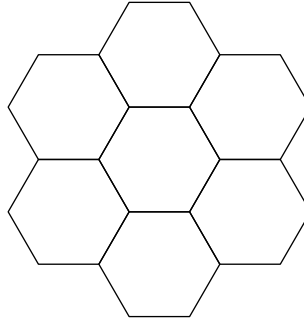
13. (15 pts) Fill in the blanks with the digits 1, 2, 3, and 4 such that no two digits in the same row, column, or 2×2 box are the same.

1	2		-	-
-	-		-	-
-	-		-	-
3	-		-	4

Let the digits in the third column from the left be a, b, c , and d from top to bottom.

Compute $125a + 25b + 5c + d$.

14. (15 pts) A circle has area 10. Another shape has twice the perimeter of the circle. What is the smallest positive integer that can't be the area of this other shape?
15. (16 pts) How many distinct subsets of the 7 hexagons can be colored black such that there are exactly 6 unordered pairs of adjacent black hexagons?



16. (16 pts) Call a positive integer *steven* if the sum of its digits is even. Compute the 2026th smallest *steven* positive integer.
- TB. (Tiebreak) Shubham loves chess, but only has the black pieces. Therefore, he has a chessboard, which is a rectangular grid with 8 squares on each side, and 16 black chess pieces. Assuming all black chess pieces are indistinguishable and each of the 64 squares on the chessboard has at most 1 piece, estimate the number of ways Shubham can place the 16 black chess pieces on the chessboard such that each of the 8 rows and each of the 8 columns has exactly 2 black pieces in it.

2026 Chantilly Math Competition - Answer Sheet
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Name: _____ Grade: _____

Email: _____

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

TB